INSTITUTE OF ENGINEERING & TECHNOLOGY (IET)
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MEDICAL INSURANCE COST PREDICTION & ANALYTICS DASHBOARD

Comprehensive Project Report & Viva Voce Examination Guide

Degree Program:	B.Tech IV Year (Final Year)
Branch:	Electronics & Telecommunication Engineering (E&TC;)
Academic Session:	2026 – 2027
Section / Group:	Section B | Group No. 11
Domain:	Software / Predictive Machine Learning
Architecture:	100% Pure Python Stack (Scikit-Learn, Streamlit, Plotly)

Prepared for: Final Year Major Project Evaluation, External Examiners, & Viva Voce Examination

1. Executive Summary & Academic Alignment

Project Abstract: Medical expenses and insurance costs vary significantly from individual to individual due to demographic characteristics, lifestyle factors, and physiological risk markers. Estimating potential insurance expenses accurately supports financial risk assessment and informed decision-making. This project develops an end-to-end machine learning system that formulates insurance cost estimation as a supervised continuous regression problem, benchmarks three algorithms on held-out test data, and delivers an interactive presentation dashboard featuring real-time predictions, exploratory data analytics, and what-if simulations.

Alignment with Academic Proforma-I Objectives:

- **Objective 1 (Predictive System):** Developed an ML regression engine capable of estimating individualized annual charges from 6 demographic/lifestyle features.
- **Objective 2 (Feature Identification):** Quantified relationships between input variables and billed charges (identifying smoking and BMI as primary drivers).
- **Objective 3 (Categorical Encoding):** Encapsulated One-Hot Encoding within Scikit-Learn ColumnTransformer pipelines with strict leakage prevention.
- **Objective 4 (Multi-Model Comparison):** Trained and benchmarked Multiple Linear Regression, Random Forest Regressor, and Gradient Boosting Regressor.
- **Objective 5 (Quantitative Evaluation):** Evaluated held-out test data using standard metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.
- **Objective 6 (Interactive Presentation App):** Delivered a 5-module responsive Streamlit dashboard with interactive Plotly visualizations and scenario analysis.

2. Dataset Specification & Statistical Audit

The project utilizes the verified **Medical Insurance Cost Dataset** (1,338 records, 7 columns) sourced from Kaggle. All 7 columns were verified through an automated data audit engine.

Feature	Data Type	Role	Observed Domain	Clinical / Demographic Significance
age	Integer	Input	18 – 64 yrs	Biological age; exhibits positive linear cost trend.
sex	Categorical	Input	female, male	Biological sex; minor overall direct effect.
bmi	Float	Input	15.96 – 53.13	Body Mass Index; non-linear threshold at BMI ≥ 30.
children	Integer	Input	0 – 5	Dependent count covered under policy.
smoker	Categorical	Input	yes, no	Tobacco smoking; single largest cost multiplier (~3.8x).
region	Categorical	Input	NE, NW, SE, SW	US residential territory; southeast shows slightly higher charges.
charges	Float (Target)	Target	$1,121.87 – $63,770.43	Billed annual hospital/insurance charge in USD.

3. Technical Architecture & Machine Learning Methodology

Strict Data Leakage Prevention Protocol:

1. **Pre-Split Partitioning:** The raw dataset was partitioned into 80% training (N=1,070) and 20% test (N=268) using a fixed random seed (seed=42) *prior* to computing any scalers, statistics, or encoders.

2. **Encapsulated Pipeline:** All preprocessing transformations are encapsulated inside Scikit-Learn ColumnTransformer objects. Scalers (StandardScaler) and categorical encoders (OneHotEncoder) are fitted strictly on X_train and serialized together with the estimators into .joblib artifacts.
3. **Zero Target Contamination:** The target column (charges) is never passed into the feature transformer.

Presentation UI	app.py pages/*.py	Streamlit multi-page interface; renders input forms, Plotly charts, KPI cards, and what-if comparisons.
Services Layer	PredictionService ScenarioService	Authoritative business logic decoupled from UI widgets; handles validation, inference, and delta calculations.
ML Pipeline	preprocessing.py registry.py	Encapsulates ColumnTransformer, loads fitted .joblib models, checks metadata.
Data Layer	loader.py validation.py	Loads data/insurance.csv, executes automated 7-column audits, checks duplicates and nulls.

4. Empirical Model Benchmarks & Results

All three models were trained on 80% data (N=1,070) and evaluated on the exact same held-out test split (20%, N=268, seed=42). The empirical results are summarized below:

Model Architecture	MAE ($ USD)	RMSE ($ USD)	R^2 Score	Rank & Verdict
Gradient Boosting Regressor	$2,401.61	$4,325.64	0.8795	■ 1st Place (Top Performer - Default)
Random Forest Regressor	$2,552.04	$4,590.28	0.8643	■ 2nd Place (Strong Non-Linear Fit)
Multiple Linear Regression	$4,181.19	$5,796.28	0.7836	■ 3rd Place (Interpretable Baseline)

Key Analytical Insights to Present to Examiners:

- **Why Gradient Boosting Won:** Healthcare costs exhibit strong synergistic interactions—specifically, patients with BMI ≥ 30 who also smoke experience exponential cost increases (> $40,000). Linear regression cannot capture this interaction without explicit polynomial interaction terms, whereas Gradient Boosting models it naturally via sequential tree splits, achieving an R^2 of 0.8795.
- **Smoking Factor Multiplier:** Regular smoking is the dominant feature. The mean cost for smokers is $32,050 compared to $8,434 for non-smokers (~3.8x higher).
- **Age Progression Slope:** Across all demographic subgroups, age exhibits a steady linear slope of approximately +$250 to +$300 per additional year of age.
- **Residual Symmetry:** Residual error diagnostics reveal zero-mean error distribution centered at $0, confirming unbiased regression estimates.

5. Interactive Presentation Dashboard Modules

app.py (Main Landing): Displays institutional header (IET DAVV B.Tech IV Year Group 11), academic disclaimer, live dataset health KPI cards (1,338 records, 100% clean), and quick navigation links.

Module A: 1_Overview.py: Comprehensive EDA with dynamic filters (Smoker/Non-smoker, Sex, Region), charges distribution with median reference line, continuous Age/BMI histograms, categorical box plots, and Pearson correlation heatmaps.

Module B: 2_Predict.py: Interactive form with sliders/dropdowns, BMI health classification indicator, real-time USD estimate card, population mean/median comparison benchmarks, multi-model estimate comparison, and JSON export.

Module C: 3_Model_Insights.py: Quantitative model benchmark table, comparative MAE/R^2 bar charts, actual vs. predicted scatter plots (y=x line), residual distribution histogram, relative feature importance chart, and full JSON benchmark export.

Module D: 4_What_If.py: Scenario simulator comparing Baseline vs. Alternative profiles with one-click presets (Stop Smoking, Reduce BMI, 10 Years Older), absolute dollar delta ($Δ), relative percent change (%Δ), and changed feature callouts.

Module E: 5_About.py: Academic proforma details, formal objectives, methodology workflow, and Kaggle open-source citations.

6. High-Yield Viva Voce Questions & Model Answers

Q1: Why is medical insurance cost prediction formulated as a regression problem instead of classification?

Answer: The target variable (charges) is a continuous numerical quantity measured in US Dollars ($ USD) ranging from $1,121.87 to $63,770.43, rather than a discrete categorical label (e.g., high/low risk). Therefore, it requires supervised regression algorithms capable of predicting real continuous values.

Q2: What is Data Leakage, and how did your project strictly prevent it?

Answer: Data leakage occurs when information from outside the training dataset (specifically from the test set) is inadvertently used during training or transformation fitting. In our project: (1) The dataset was partitioned into 80% train and 20% test *before* any feature scaling or encoding. (2) All scalers (StandardScaler) and encoders (OneHotEncoder) were fitted *strictly on X_train* inside Scikit-Learn ColumnTransformer pipelines. (3) The test set was held out exclusively for final unbiased evaluation.

Q3: Why did you use One-Hot Encoding instead of Label Encoding for categorical features?

Answer: Categorical features such as region (northeast, northwest, southeast, southwest) and sex have no natural ordinal hierarchy. Label encoding assigns arbitrary integers (e.g., 1, 2, 3, 4), which would falsely imply that southwest (4) has a greater magnitude than northeast (1), distorting linear regression models. One-Hot Encoding creates independent binary indicator columns without artificial ordering.

Q4: Why did Gradient Boosting and Random Forest significantly outperform Linear Regression?

Answer: Multiple Linear Regression assumes a strictly additive linear equation. However, medical costs exhibit severe non-linear threshold effects and synergistic interactions. For example, individuals with BMI ≥ 30 who also smoke experience exponential cost increases (> $40,000) rather than a simple sum of BMI and smoking penalties. Decision tree ensembles naturally segment feature spaces and capture non-linear interactions, achieving a much higher R^2 (0.8795 vs. 0.7836).

Q5: What is the mathematical difference between MAE and RMSE, and why is RMSE notably higher here?

Answer: MAE is the average absolute error ($\frac{1}{n}\sum |y_i - \hat{y}_i|$), giving linear weight to all deviations. RMSE squares errors before averaging ($\sqrt{\frac{1}{n}\sum (y_i - \hat{y}_i)^2}$), heavily penalizing large outliers. In medical billing data, rare extreme hospitalization claims (e.g. $60,000) create larger residual penalties, which naturally causes RMSE ($4,325) to be higher than MAE ($2,401).

Q6: What does the R^2 score of 0.8795 mean in practical terms? Can R^2 be negative?

Answer: An R^2 score of 0.8795 means our Gradient Boosting model explains approximately 87.95% of the total variance observed in annual medical charges. Yes, R^2 can be negative on an unseen test set if the model makes worse predictions than a naive baseline that simply predicts the mean target value.

Q7: Why does the application include an explicit 'Non-Causal' disclaimer?

Answer: The Kaggle dataset is historical and observational, not derived from a randomized clinical trial. While smoking correlates strongly with higher predicted expenses, machine learning regression models describe statistical correlations, not certified medical interventions or guaranteed financial quotes.

Q8: How did you ensure software engineering quality and test coverage in this project?

Answer: We built a comprehensive automated test suite with 21 unit and integration tests using pytest, covering schema validation, out-of-bounds boundary warnings, data auditing, pipeline fitting, metric formulas, and scenario delta calculations with 100% passing status.

7. Final Presentation Checklist & Viva Tips

- ✓ **Opening Demo Flow:** Start on app.py to explain project objectives and dataset quality (1,338 records, 0 missing). Navigate to 1_Overview.py to show the smoking cost multiplier and BMI threshold.
- ✓ **Live Prediction:** Move to 2_Predict.py, enter a sample profile (e.g. Age 40, BMI 33, Smoker=Yes), and show the estimated cost ($38k+) along with the population benchmark and multi-model comparison.
- ✓ **What-If Simulator (Examiner Favorite):** Go to 4_What_If.py and click 'Stop Smoking Scenario'. Point out how charges decrease by ~$20k-$25k and explain the absolute and percentage delta metrics.
- ✓ **Model Justification:** When asked which model is best, refer to 3_Model_Insights.py: state that Gradient Boosting achieved lowest MAE ($2,401) and highest R^2 (0.8795), and show the residual plot proving unbiased errors.
- ✓ **Testing & Code Quality:** Mention that all 21 automated pytest test cases pass cleanly, demonstrating software engineering rigor.